Sharp finite-size coexistence in an adaptive multi-species competitive network: scaling evidence for a continuous transition

Evan W. Martin

Independent researcher,

Abstract

A three-colour competitive model is studied on a fixed toroidal lattice and on an adaptive network where edges co-evolve with node states. On the fixed lattice the transition is smooth. On the adaptive network the same local rules produce an unusually sharp transition: boundary density jumps from $\rho_b \approx 0.17$ to ≈ 0.48 across $\mu \in [0.35, 0.355]$, distributions are bimodal at intermediate system sizes ($L = 128, 256$), and a variance spike rises four to five orders of magnitude above the quiet-phase background. Across $L = 128 \to 256 \to 384$ the Binder cumulant dip shallows monotonically toward 2/3 and the bimodal window narrows, consistent with a sharp continuous transition over the tested size range; a weakly first-order alternative with crossover length exceeding $L = 384$ cannot be excluded. A degree-cap sweep ($k_{\max} \in \{4, 5, 6, 8\}$) implicates degree headroom as a contributing topological ingredient: capping at $k_{\max} = 4$ yields smooth behaviour, while every $k_{\max} \geq 5$ yields a sharp jump whose location drifts toward higher μ with increasing headroom. The headroom reading is partially confounded with a reinforcement-firing-rate effect (at $k_{\max} = 4$ the mean degree falls below the firing threshold) and, in the local-only control, with candidate-pool shrinkage, so non-local formation per se is implicated only indirectly.

1. Introduction

Multi-species competition on lattices is a well-studied route to spatial pattern formation. Models of cyclic and non-cyclic competition produce phase transitions between ordered (species-segregated) and disordered (well-mixed)

states whose universality depends on lattice geometry, number of species, and the form of the local dynamics [1, 2, 3]. On fixed regular lattices, these transitions are typically continuous.

Real populations, however, do not interact on static topologies. Individuals form and break social, spatial, or ecological links in response to their own state and the states of their neighbours—a co-evolutionary feedback between dynamics *on* the network and dynamics *of* the network [4]. A growing body of work shows that this feedback can qualitatively change collective behaviour, including the order of phase transitions. Holme and Newman [5] and Vazquez et al. [6] demonstrated that adaptive rewiring converts the continuous fragmentation transition in the voter model to a discontinuous one. Gross et al. [7] found analogous sharpening in adaptive epidemic models, and Demirel et al. [8] studied adaptive networks with cyclic dominance, showing that coevolution of topology and strategy leads to spontaneous network fragmentation.

Here we ask whether topological co-evolution changes the character of the phase transition in a *conviction-weighted* multi-species competition model, and if so, what specific feature of the rewiring drives the change. Our system differs from prior adaptive-network studies in three respects: (i) node states are continuous three-channel vectors (not discrete opinions or strategies), giving a notion of "conviction" that modulates interaction outcomes; (ii) rewiring is globally non-local—new edges may form between any two compatible nodes anywhere in the system, subject to a degree cap—rather than being restricted to neighbours of neighbours; (iii) we compare directly against a fixed-lattice control to assess the effect of topological freedom.

We find that the same local rules that produce a smooth transition on the fixed lattice produce an unusually sharp transition on the adaptive network: a jump in coupled order parameters, bimodal coexistence visible in histograms at intermediate system sizes, and a variance spike. However, the Binder cumulant dip shallows monotonically from $L = 128$ to $L = 384$, consistent with a sharp but continuous transition over the tested size range. A degree-cap sweep ($k_{\max} \in \{4, 5, 6, 8\}$) implicates degree headroom as a contributing topological ingredient, with the jump location drifting monotonically toward higher μ as headroom grows. The headroom reading is partially confounded with a reinforcement-firing-rate effect at $k_{\max} = 4$ and, in the local-only control, with candidate-pool shrinkage; non-local formation per se is implicated only indirectly.

The remainder of the paper is organised as follows. Section 2 defines the

model. Sections 3 and 4 present results on the fixed lattice and adaptive network, respectively, including control experiments. Section 5 gives finite-size scaling evidence. Section 6 discusses the mechanism and connections to related models.

2. Model

The model is an interacting particle system on an adaptive network: nodes carry continuous $[0,1]^3$ state vectors, updates are asynchronous and random-sequential, and in the adaptive variant the topology co-evolves with the dynamics. We use "cellular automaton" loosely, following the convention in lattice competition literature.

2.1. Fixed lattice

The substrate is an $L \times L$ two-dimensional torus: a square grid with periodic boundary conditions in both directions, giving every node exactly four neighbours. This 4-regular toroidal lattice serves as the fixed-topology control.

2.2. Node state and competition

Each node i carries a three-channel colour value $\boldsymbol{c}_i = (r_i, g_i, b_i) \in [0,1]^3$. The three channels represent three competing species; the *dominant type* of node i is the channel with the largest value. Ties are broken by a fixed cyclic order $R > G > B > R$. (This symmetry-breaking is arbitrary; the three species are otherwise identical, and the phase diagram is symmetric under relabelling.) The continuous colour values endow each node with a "conviction": the *dominance margin* is the difference between the dominant channel and the maximum of the other two. A node with colour $(0.78, 0.12, 0.04)$ is strongly red (margin 0.67), while $(0.35, 0.33, 0.31)$ is weakly red (margin 0.02) and easily converted.

At each elementary step a node i is chosen uniformly at random and interacts with each of its neighbours j. Channel values change in increments of $\delta = 1/255 \approx 0.004$:

- **Agreement** (same dominant type): both i and j increment their shared dominant channel by δ (clamped at 1).

- **Conflict** (different dominant types): the node with the smaller dominance margin shifts toward the winner's type: the winner's channel is incremented by δ and the loser's own dominant channel is decremented by δ. When margins are equal, the focal node i concedes.

A collective reinforcement rule amplifies local consensus: if at least $r_{\min}$ of i's neighbours share the same dominant type (as determined by the edge relationship after competition), that channel receives an additional increment. All experiments use $r_{\min} = 4$ unless otherwise noted.

One *frame* consists of $N = L^2$ elementary steps, corresponding to one sweep of the lattice in expectation.

2.3. Mutation

After node i has interacted with all its neighbours and reinforcement has been applied, i is reset to a uniformly random colour—each channel drawn independently and uniformly from $[0, 1]$—with probability μ. Mutation is applied once per node activation, not once per pairwise interaction. This mutation is the noise parameter that drives the phase transition: low μ favours local order; high μ disrupts it.

2.4. Adaptive topology

The adaptive variant begins from the same 4-regular toroidal lattice but allows the network to rewire. After each pairwise interaction between nodes i and j:

- If the edge is **compatible** (endpoints share dominant type after competition): with probability $1/N$, a new edge is created between node i and a uniformly random compatible node anywhere in the system, provided both endpoints are below the degree cap $k_{\max}$. Up to 8 random candidates are tried; if none qualifies (due to degree cap, existing edge, or self-loop), the attempt is discarded.

- If the edge is **incompatible** (endpoints differ): with probability $1/N$, the edge is severed, provided both endpoints retain degree ≥ 2.

The maximum degree is capped at $k_{\max} = 8$ (twice the initial lattice degree; the qualitative results are not sensitive to this choice provided $k_{\max}$ is moderately above 4, as it determines the dynamic range of $\langle k \rangle$). The minimum degree floor of 2 reduces the frequency of isolated nodes and extreme fragmentation, but does not guarantee global connectivity.

Since each of N node activations per frame involves $O(k)$ pairwise interactions, each with rewiring probability $1/N$, the expected number of rewiring events per frame is $O(\langle k \rangle)$—of order unity per frame, scaling with the system.

In the ordered phase ($\mu < \mu_c$), nodes cluster near the degree cap: at $\mu = 0.33$, approximately 82% of nodes have $k = 8$ and 95% have $k \geq 7$. In the disordered phase ($\mu > \mu_c$), the distribution shifts toward the floor: at $\mu = 0.50$, approximately 44% of nodes have $k = 2$ and 71% have $k \leq 3$. The degree distribution is thus strongly phase-dependent (Fig. 2, Table 3).

The result is that ordered regions recruit long-range connections while disordered boundaries shed them. The topology and dynamics co-evolve: the network is simultaneously the arena for competition and a product of it. A control variant restricts new-edge formation to nodes within graph distance 2 of the focal node while keeping all other rules identical (Section 4.4).

2.5. Algorithm summary

Table 1 gives the per-node update. One frame repeats this N times with uniformly random node selection.

Table 1: Per-node update for node i.

```
Snapshot neighbour list N(i)
for each j ∈ N(i):
    if same dominant type:
        increment shared channel on both i and j
        [adaptive] with prob. 1/N: form edge to random
        compatible node
    else:
        weaker node shifts one unit toward winner
        [adaptive] with prob. 1/N: sever edge i–j
        if both keep degree ≥ 2
Reinforce: if ≥ r_min edges agree, boost that type
Mutate: with prob. μ, reset c_i to random
```

2.6. Observables

The primary order parameter is the *boundary density* ρ_b: the fraction of edges connecting nodes of different dominant types. Low ρ_b ($\lesssim 0.15$)

indicates species-segregated order; high ρ_b ($\gtrsim 0.5$) indicates a well-mixed disordered state.

On the adaptive network, the mean degree $\langle k \rangle$ serves as a second, coupled order parameter: $\langle k \rangle \approx 7$–$8$ in the ordered phase (near the cap) and $\langle k \rangle \approx 3$–$4$ in the disordered phase (near the floor).

All reported values are time-averages over the second half of each run (discarding transients). Error bars on mean quantities in figures show the standard error of the mean ($\sigma/\sqrt{n}$) across independent seeds. Tables report the seed-to-seed standard deviation σ and population variance. The primary experiments use 64 seeds; the Binder cumulant analysis uses 64 seeds per μ point at every system size, including $L = 384$.

2.7. Note on the reinforcement threshold

On the fixed 4-regular lattice, $r_{\min} = 4$ requires all four neighbours to agree before reinforcement fires—effectively a unanimity rule. On the adaptive network, where degree can reach $k_{\max} = 8$, the same $r_{\min} = 4$ requires only half the neighbours to agree. This asymmetry is inherent to applying a fixed threshold across variable-degree topologies. We address its consequences with a control experiment (Section 4.5) showing that varying $r_{\min}$ on the fixed lattice does not change the transition character.

3. Fixed lattice: smooth transition

On the fixed 4-regular torus ($L = 128$, 64 seeds, 50 000 frames), boundary density decreases smoothly with decreasing mutation probability (Fig. 1, Table 2). No discontinuity is observed at any μ tested: ρ_b ranges from 0.48 at $\mu = 0.33$ to 0.08 at $\mu = 0.10$, with a smooth crossover near $\mu \approx 0.20$. This section serves as a baseline: the fixed lattice transition is smooth, and Section 4 asks what changes under adaptive rewiring.

4. Adaptive network: sharp transition

Allowing edges to co-evolve qualitatively changes the transition.

4.1. Sharp jump in coupled order parameters

The phase diagram of the adaptive network (Fig. 2, Table 3) shows both order parameters changing sharply between $\mu = 0.35$ and $\mu = 0.355$. At $\mu = 0.35$, the majority of seeds converge to the ordered phase ($\rho_b \approx 0.17$,

Table 2: Fixed lattice phase diagram (128×128, $r_{\min} = 4$, 64 seeds, 50k frames).

μ	$\bar{\rho}_b$	$\sigma(\rho_b)$	$\mathrm{var}(\rho_b)$
0.1000	0.0846	0.0059	3.42e-05
0.1111	0.0958	0.0074	5.42e-05
0.1250	0.1076	0.0066	4.31e-05
0.1429	0.1255	0.0035	1.20e-05
0.1667	0.1577	0.0036	1.29e-05
0.2000	0.2284	0.0016	2.66e-06
0.2500	0.3883	0.0012	1.39e-06
0.3333	0.4766	0.0008	7.04e-07

$\langle k \rangle \approx 6.0$); at $\mu = 0.355$, most seeds have crossed to the disordered phase ($\rho_b \approx 0.48$, $\langle k \rangle \approx 4.2$). By $\mu = 0.36$, all seeds are fully disordered ($\rho_b = 0.55$, $\langle k \rangle = 3.9$). The boundary density increases by a factor of $\sim$3× and the mean degree drops by $\sim$1.5× across this narrow window.

4.2. Topology-independent confirmation and coexistence

Because ρ_b and $\langle k \rangle$ depend on the edge set, the sharp jump could in principle be a denominator artifact. A purely state-based observable, the dominant-species fraction $f_{\mathrm{dom}} = \max(n_R, n_G, n_B)/N$ (Fig. 3), drops sharply from ≈ 0.9 to $\approx 1/3$ across the same window, indicating a genuine state transition rather than a denominator artifact.

Histograms of per-seed ρ_b across the transition (Fig. 4) reveal the distribution shape more precisely than the scatter plots above. At $\mu = 0.33$, the distribution is a tight unimodal peak near $\rho_b \approx 0.10$. At $\mu = 0.345$, it remains unimodal but broadens. At $\mu = 0.35$, the distribution develops a long right tail, with most seeds near $\rho_b \approx 0.16$ but a few reaching $\rho_b > 0.3$. At $\mu = 0.355$, the distribution is *clearly bimodal*: one cluster near $\rho_b \approx 0.15$ and a second near $\rho_b \approx 0.55$, separated by a gap with no seeds. By $\mu = 0.37$, all seeds have collapsed to a tight unimodal peak in the disordered phase.

The bimodality at $\mu = 0.355$—with two well-separated peaks and an empty gap—is the strongest coexistence-like signature at $L = 128$. Section 5.1 shows that these signatures weaken with increasing system size.

4.3. Variance spike and initial-condition dependence

The seed-to-seed variance of ρ_b peaks at $\mu = 0.355$ with $\mathrm{var}(\rho_b) = 1.5 \times 10^{-2}$—four to five orders of magnitude above the quiet-phase background

Table 3: Adaptive network phase diagram (128×128, $r_{\min} = 4$, $k_{\max} = 8$, 64 seeds, 30k frames).

μ	$\bar{\rho}_b$	$\sigma(\rho_b)$	$\mathrm{var}(\rho_b)$	$\langle k \rangle$
0.1111	0.0270	0.0002	5.82e-08	7.90
0.1250	0.0306	0.0002	5.31e-08	7.89
0.1429	0.0352	0.0002	6.23e-08	7.88
0.1667	0.0418	0.0003	7.62e-08	7.87
0.2000	0.0511	0.0003	1.01e-07	7.85
0.2500	0.0665	0.0004	1.26e-07	7.80
0.3300	0.1016	0.0012	1.36e-06	7.35
0.3333	0.1046	0.0016	2.51e-06	7.29
0.3400	0.1131	0.0034	1.18e-05	7.06
0.3450	0.1257	0.0093	8.59e-05	6.75
0.3500	0.1748	0.0566	3.20e-03	5.97
0.3550	0.4779	0.1230	1.51e-02	4.21
0.3600	0.5483	0.0005	2.89e-07	3.90
0.3700	0.5511	0.0007	4.34e-07	3.87
0.4000	0.5589	0.0007	5.03e-07	3.79
0.4500	0.5706	0.0007	5.25e-07	3.68
0.5000	0.5813	0.0007	5.41e-07	3.59
1.0000	0.6581	0.0006	3.66e-07	3.12

(Fig. 2, inset).

To test for initial-condition dependence, we run at $\mu = 0.34$, 0.35, and 0.36 from both random and spatially ordered initial conditions (64 seeds each; Table 4). At $\mu = 0.34$, both conditions converge to order; at $\mu = 0.36$, both converge to disorder. At $\mu = 0.35$, a small fraction of seeds escape to the opposite basin (3/64 random-init, 2/64 ordered-init), with means close in both cases ($\bar{\rho}_b = 0.175$ vs. 0.178). This is weak initial-condition dependence rather than strong hysteresis.

4.4. Controls

To isolate which features of the rewiring drive the sharpening, we ran two families of controls: a local-only control restricting new edges to distance-2 neighbours (all other rules identical), and a degree-cap sweep varying $k_{\max}$.

Figure 5 compares four systems at $L = 128$ (64 seeds, 30 000 frames): the

Table 4: Initial-condition dependence across the bistable window (128×128, 64 seeds, 30k frames).

μ	Init	$\bar{\rho}_b$	$\sigma(\rho_b)$	$\langle k \rangle$
0.340	random	0.1131	0.0034	7.06
	ordered	0.1133	0.0036	7.06
0.345	random	0.1257	0.0093	6.75
	ordered	0.1238	0.0075	6.80
0.350	random	0.1748	0.0566	5.97
	ordered	0.1778	0.0515	5.95
0.355	random	0.4779	0.1230	4.21
	ordered	0.4458	0.1438	4.36
0.360	random	0.5483	0.0005	3.90
	ordered	0.5482	0.0007	3.90

fixed lattice, the local-only adaptive network ($k_{\max} = 8$), the global adaptive network with $k_{\max} = 4$ (matching the fixed-lattice degree), and the standard global adaptive network with $k_{\max} = 8$.

On the local-only network, ρ_b rises smoothly from ≈ 0.13 at $\mu = 0.33$ to ≈ 0.48 at $\mu = 0.50$, with no sharp jump at any μ tested. The restriction removes the sharp transition, but it also shrinks the candidate pool from $O(N) \approx 16{,}000$ compatible nodes system-wide to $O(k^2) \approx 50$–60 within graph distance 2 at $k_{\max} = 8$, so it does not isolate non-locality from formation rate.

A second family of controls varies the degree cap. At $k_{\max} = 4$ on the global adaptive network (matching the fixed-lattice degree), the sharp transition vanishes: ρ_b rises smoothly from ≈ 0.15 ($\mu = 0.15$) to ≈ 0.56 ($\mu = 0.50$), with seed-to-seed variance three orders of magnitude below the $k_{\max} = 8$ case. Because the network starts at degree $4 = k_{\max}$, there is no headroom for edge accumulation: at initialisation every node is already at the cap, so formation attempts fail until severance opens a slot, and mean degree drops monotonically to $\langle k \rangle \approx 3.0$–3.7.

To disentangle headroom from outright formation suppression, we ran intermediate-headroom controls at $k_{\max} = 5$ and $k_{\max} = 6$ (one and two units of headroom respectively). Both recover a sharp coexistence-like jump

Table 5: Finite-size scaling at $\mu = 0.35$ (32 seeds per condition).

L	Init	$\bar{\rho}_b$	$\sigma(\rho_b)$	$\langle k \rangle$
64	random	0.1004	0.0008	7.80
	ordered	0.1006	0.0009	7.80
128	random	0.1819	0.0705	5.92
	ordered	0.1700	0.0500	6.07
256	random	0.4637	0.1151	4.21
	ordered	0.4517	0.1246	4.25

(Fig. 5): at $k_{\max} = 5$, ρ_b jumps from ≈ 0.20 at $\mu = 0.27$ to ≈ 0.51 at $\mu = 0.30$; at $k_{\max} = 6$, the jump shifts to between $\mu = 0.30$ and $\mu = 0.33$ (from $\rho_b \approx 0.14$ to ≈ 0.53); at $k_{\max} = 8$, the jump occurs near $\mu \approx 0.345$. The transition location drifts monotonically toward higher μ with increasing headroom. Over the tested values, the sharp jump is absent at $k_{\max} = 4$ and present for $k_{\max} \geq 5$, suggesting a threshold-like onset of the sharp coexistence-like behaviour once some degree headroom is available.

Together with the fixed-lattice baseline and the local-only control, this $k_{\max}$ sweep implicates degree headroom as a contributing topological ingredient: the sharp behaviour requires some room for like-state edges to accumulate, but does not require the full $k_{\max} = 8$ headroom of the standard setting. The headroom interpretation is partially confounded with a reinforcement-firing-rate effect at $k_{\max} = 4$ (see Section 6.1), and the local-only control is confounded by candidate-pool shrinkage; an unconfounded test of either would require an $r_{\min}$ sweep on the adaptive network or a matched-formation-rate non-local control respectively, both left to future work.

4.5. Reinforcement-threshold control

To rule out an effective-threshold artifact ($r_{\min} = 4$ is unanimity at degree 4 but only a 50% majority at degree 8), we repeated the fixed-lattice experiment with $r_{\min} = 2$ (64 seeds, 50 000 frames), matching the fractional threshold of $r_{\min} = 4$ at degree 8. The transition remains smooth and ρ_b values shift downward only slightly, making effective-threshold mismatch unlikely as the explanation.

5. Finite-size scaling

We measure ρ_b at $\mu = 0.35$ for $L = 64$, 128, and 256, with both random and ordered initial conditions (32 seeds each; Table 5).

At $L = 64$, all 64 seeds converge tightly to the ordered phase ($\rho_b \approx 0.10$, $\langle k \rangle \approx 7.8$, combined $\mathrm{var}(\rho_b) = 7.1 \times 10^{-7}$). At $L = 128$, seeds split between basins with $\mathrm{var}(\rho_b) = 3.8 \times 10^{-3}$. At $L = 256$ (100 000 frames for equilibration), most seeds have crossed to the disordered side ($\bar{\rho}_b \approx 0.46$) but variance remains high ($\mathrm{var}(\rho_b) = 1.4 \times 10^{-2}$), indicating that the system is still bistable at this size.

The combined variance *increases* by four orders of magnitude from $L = 64$ to $L = 256$. At $\mu = 0.35$ the pseudocritical region drifts toward this value with increasing L (Section 5.1), so part of the variance increase reflects the approach to the pseudocritical point. Nonetheless, the strong broadening and bimodal splitting at $L = 128$ and 256 indicate coexistence-like behaviour at accessible system sizes.

5.1. Binder cumulant

The Binder cumulant $U_L = 1 - \langle \rho_b^4 \rangle / (3 \langle \rho_b^2 \rangle^2)$, where averages are over seeds, provides a standard test for first-order transitions: $U_L \to 2/3$ for a single-peaked distribution, and U_L develops a negative dip when the distribution is bimodal.

Figure 6 shows U_L vs. μ for $L = 64$, 128, 256, and 384, all with fine μ grids ($\Delta\mu = 0.001$; 64 seeds at every size). Run lengths scale with system size: $L = 64$ and 128 use 30 000–50 000 frames, $L = 256$ uses 100 000 frames, and $L = 384$ uses 200 000 frames—twice as long as $L = 256$, partially compensating for slower equilibration at the largest size. Spot-checks of ρ_b time series at $L = 384$ show similar within-run basin-switching near μ_c to that documented at $L = 256$ (Supplementary Material, Sec. S3); the doubled run length improves but does not eliminate this near-critical residence-mixing effect.

The *position* of the minimum shifts to lower μ with increasing L: $\mu^*(64) \approx 0.360$, $\mu^*(128) \approx 0.352$, $\mu^*(256) \approx 0.349$, $\mu^*(384) \approx 0.348$. The drift slows with increasing L, consistent with convergence toward a finite limiting value, though four points do not permit a reliable extrapolation. However, the dip *depth* shallows monotonically (bootstrap 95% CI in brackets): $U_{\min} = -0.39$ $[-0.85, -0.05]$ ($L = 64$), 0.26 $[0.18, 0.36]$ ($L = 128$), 0.52 $[0.49, 0.55]$ ($L = 256$), and 0.58 $[0.56, 0.61]$ ($L = 384$); all with 64 seeds. The confidence intervals do not overlap between successive sizes from $L = 128$

onward, consistent with statistically robust shallowing given the seed counts. (At $L = 64$ the system contains only 4096 nodes and the deep negative dip may partly reflect small-system fluctuations; the shallowing trend from $L = 128$ onward is the more reliable indicator.) For a standard first-order transition the dip should deepen without bound as $L \to \infty$; instead the minimum recovers steadily toward the single-phase value $U = 2/3$, consistent with a sharp but *continuous* transition over the tested size range. A weakly first-order alternative, in which the Binder crossover length exceeds $L = 384$, cannot be excluded with four sizes; settling the question definitively would require systems beyond $L = 384$.

Per-seed ρ_b distributions at $L = 384$ (Supplementary Material, Sec. S2) show that bimodality persists at the largest size but is confined to a narrow window $\mu \in [0.347, 0.349]$ around the Binder minimum—substantially narrower than the $L = 128$ bimodal window of Figure 4, consistent with the same sharpening-yet-continuous trend as the Binder shallowing.

A two-parameter Binder collapse $U_L(\mu) = \tilde{U}((\mu - \mu_c)L^{1/\nu})$ does not succeed for $L \in \{128, 256, 384\}$: the $L = 128$ dip is too deep to be absorbed by any rescaling, consistent with $L = 128$ lying outside the regime where the leading-order finite-size scaling form holds. Restricting to $L \in \{256, 384\}$ yields a tight collapse, but with only two curves and two free parameters the exponent is weakly constrained (bootstrap 95% CI: $1/\nu \in [0.97, 1.99]$, spanning standard two-dimensional continuous-transition exponents without selecting one). The collapse is therefore consistent with a continuous transition over the tested sizes but does not assign a specific universality class; see Supplementary Material, Sec. S1 for the full analysis and figure.

6. Discussion

6.1. Mechanistic interpretation

The control experiments in Section 4.4 narrow the set of topological ingredients that contribute to the sharp transition. Sweeping $k_{\max} \in \{4, 5, 6, 8\}$ shows that any degree headroom (one unit or more) recovers the sharp coexistence-like jump, with the jump location shifting monotonically toward higher μ as headroom grows. The local-only control also restores smoothness but is confounded by candidate-pool shrinkage from $O(N)$ to $O(k^2)$ and reduced effective formation rate; non-local formation per se is implicated only indirectly. Whether the formation/severance asymmetry contributes beyond

non-local formation alone has not been tested directly (the converse control, making severance also non-local, remains future work).

A second confound bounds the headroom interpretation. At $k_{\max} = 4$, mean degree in the ordered phase drops to $\langle k \rangle \approx 3.1$, below the reinforcement threshold $r_{\min} = 4$; reinforcement firing is therefore physically suppressed by construction at $k_{\max} = 4$, not by absence of a homophily–topology feedback loop alone. The $k_{\max} = 4$ vs. $k_{\max} \geq 5$ contrast partly reflects this trivial reinforcement-firing-rate effect. A more interpretable signal is the monotonic μ-shift of the jump location across $k_{\max} \in \{5, 6, 8\}$, where $\langle k \rangle \approx 4.8, 5.8, 7.9$ in the ordered phase (all above $r_{\min}$) and the jump location moves from $\mu \approx 0.30$ to 0.345 with increasing headroom. The direction of this shift is consistent with the homophily–topology picture: more headroom yields larger ordered-phase $\langle k \rangle$, so each node experiences more reinforcement firings per frame in the ordered phase, making it more robust against mutation and shifting μ_{c} upward. Even within the $k_{\max} \geq 5$ regime the effective ratio $r_{\min}/k$ varies from ≈ 0.83 to 0.51, so the shift cannot be cleanly attributed to headroom alone; a definitive disentanglement would require an $r_{\min}$ sweep on the adaptive network at fixed $k_{\max} = 8$, which we leave to future work.

The observed behaviour is consistent with a homophily–topology feedback loop: in the ordered phase, compatible neighbours accumulate degree ($\langle k \rangle \approx 7$–$8$ at $k_{\max} = 8$), providing more reinforcement and competition partners of the same type and stabilising order against mutation; in the disordered phase, incompatibility drives edge severance, lowering $\langle k \rangle$ to ≈ 3–4, and sparse connectivity makes spontaneous ordering difficult. At intermediate system sizes this loop produces coexistence-like behaviour in which ordered and disordered basins are not cleanly separated; the Binder cumulant trend (Section 5.1) shows that these signatures weaken at larger L, indicating that any effective barrier does not grow unboundedly with system size. This picture is qualitatively consistent with prior adaptive-network results in voter, SIS, and cyclic-dominance models [5, 6, 7, 9, 10, 8, 11], where positive feedback between state coherence and network structure has been identified as a generic sharpening mechanism.

6.2. Limitations and future work

Several aspects of this study would benefit from further investigation. Equilibration near μ_{c} remains the principal technical limitation: time-series spot-checks at both $L = 256$ and $L = 384$ show that some seeds undergo

within-run basin-switching during the averaging window, so second-half averages near the Binder minimum partly conflate the two basins (Supplementary Material, Sec. S3). A systematic relaxation-time measurement (frames to convergence vs. L) would clarify this further.

The Binder cumulant dip shallows monotonically from $L = 128$ to $L = 384$, consistent with a continuous transition over the tested size range. The two-parameter Binder collapse at $L \in \{256, 384\}$ is consistent with standard two-dimensional continuous-transition exponents but is weakly constrained by only two curves (see Supplementary Material, Sec. S1); a definitive universality-class assignment would require simulations at $L \geq 512$ or correction-to-scaling fits with additional free parameters.

The mechanism claim rests on ablation controls, not on a direct measurement of the feedback loop. The single highest-value missing control is an $r_{\min}$ sweep on the adaptive network at fixed $k_{\max} = 8$, which would disentangle degree headroom from the reinforcement-firing-rate effect identified in Section 6.1. A matched-formation-rate non-local control and a non-local-severance control would further sharpen the roles of formation range and the formation/severance asymmetry, respectively. A mean-field theory for the coupled ρ_b–$\langle k \rangle$ dynamics would help anchor the simulation results and predict the location of the bistable window as a function of model parameters.

7. Conclusion

On a fixed lattice the three-colour competitive model shows a smooth transition. Adaptive non-local rewiring sharpens the transition into a coexistence-like one with a jump in coupled order parameters ($\rho_b : 0.17 \to 0.48$, $\langle k \rangle : 6.0 \to 4.2$ across $\mu \in [0.35, 0.355]$), bimodal histograms at intermediate system sizes, and a variance spike four to five orders of magnitude above the quiet-phase background. Across $L = 128 \to 256 \to 384$ the Binder cumulant dip shallows monotonically and the bimodal window narrows—consistent with a sharp but continuous transition over the tested size range, with a weakly first-order alternative beyond $L = 384$ not excluded. A degree-cap sweep ($k_{\max} \in \{4, 5, 6, 8\}$) implicates degree headroom as a contributing topological ingredient, with the jump location drifting monotonically toward higher μ as headroom grows. The headroom interpretation is partially confounded with a reinforcement-firing-rate effect at $k_{\max} = 4$ (mean degree falls below $r_{\min}$) and, in the local-only control, with candidate-pool shrinkage; a

definitive disentanglement would require an $r_{\min}$ sweep on the adaptive network at fixed $k_{\max} = 8$.

CRediT authorship contribution statement

Evan W. Martin: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code, raw per-seed data, and analysis scripts that support the findings of this study are publicly available at `https://github.com/evanwmart/competitive-ca`.

Acknowledgments

The author declares no external funding. Simulations were performed on personal computing hardware.

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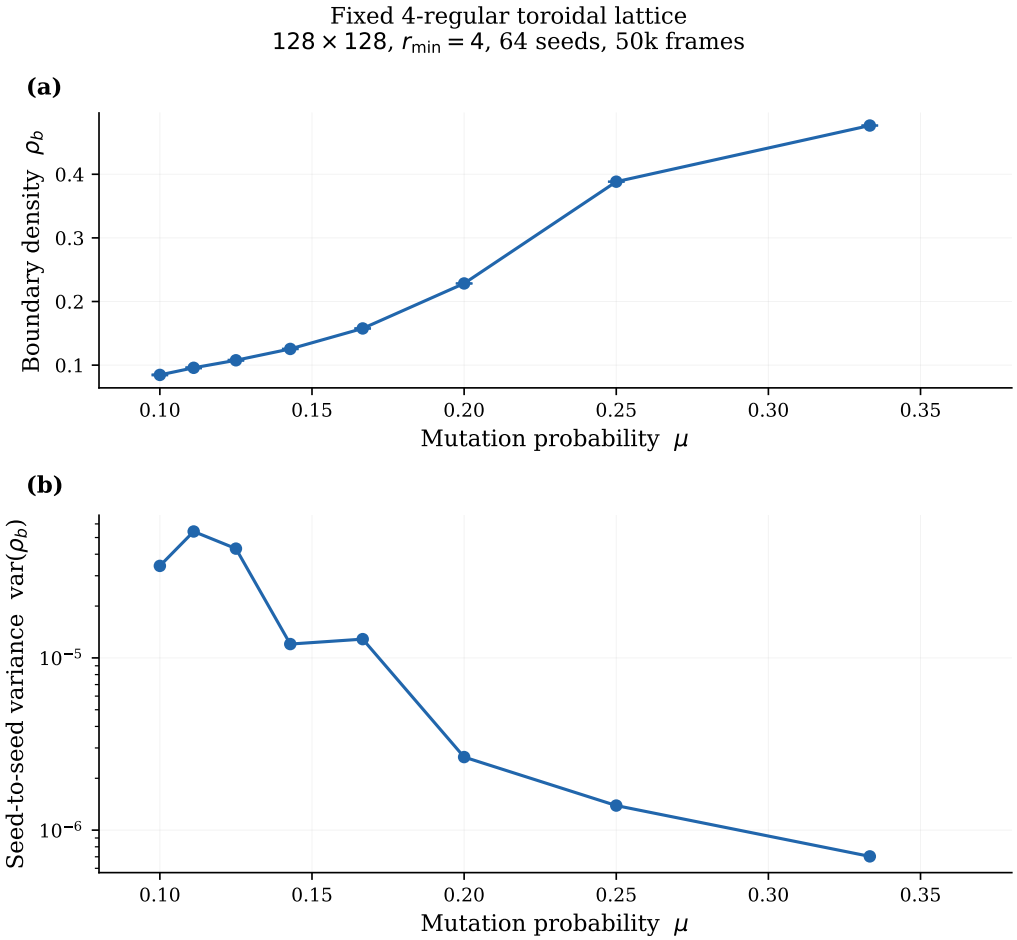

Figure 1: **Fixed lattice phase diagram** (128×128, $r_{\min} = 4$, 64 seeds, 50k frames). (a) Mean boundary density ρ_b vs. μ. The transition is smooth. (b) Seed-to-seed variance of ρ_b.

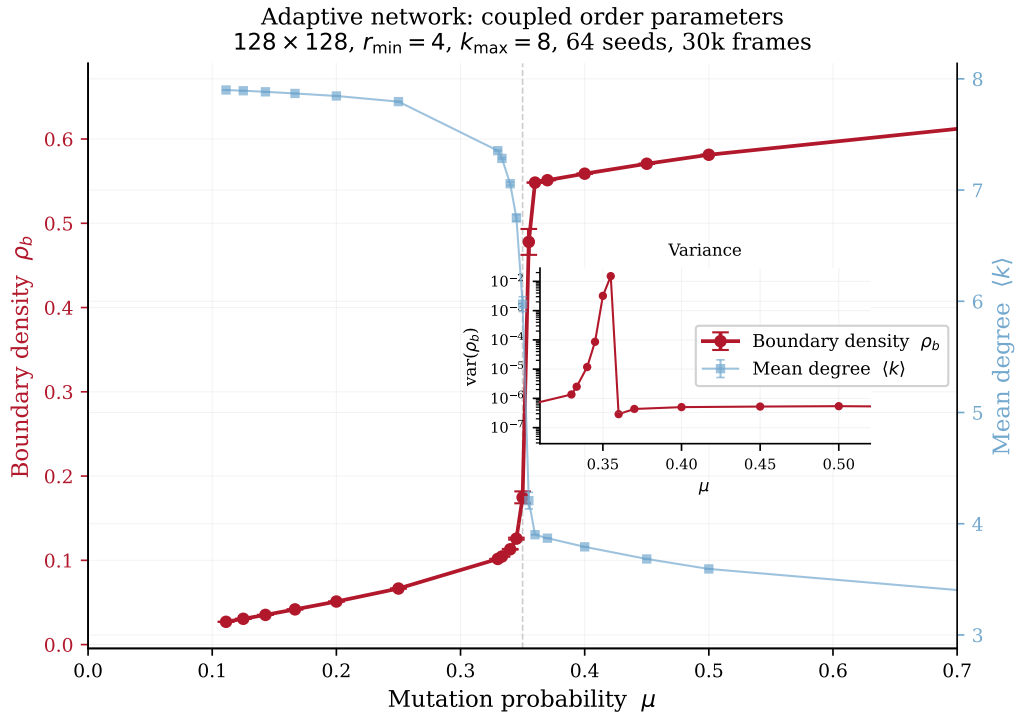

Figure 2: **Adaptive network phase diagram** (128×128, $r_{\min} = 4$, $k_{\max} = 8$, 64 seeds, 30k frames). Boundary density ρ_b (red, left axis) and mean degree $\langle k \rangle$ (blue, right axis) vs. mutation probability μ. Both order parameters change sharply in the window $\mu \in [0.35, 0.355]$.

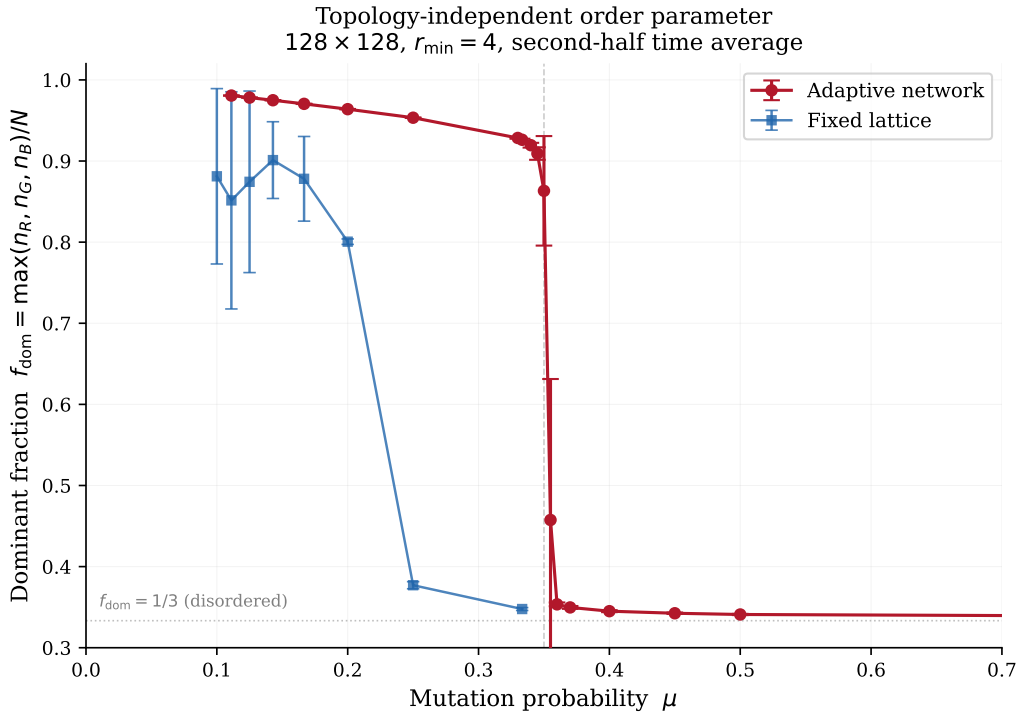

Figure 3: **Topology-independent order parameter.** f_{dom} vs. μ for the adaptive network (red) and fixed lattice (blue). The sharp drop indicates that the transition reflects a genuine state change. Dashed line: $f_{\text{dom}} = 1/3$ (disordered limit).

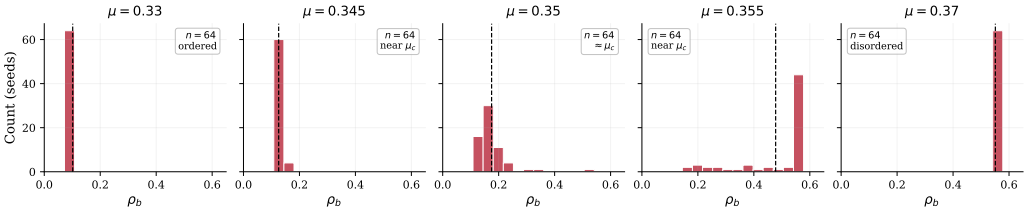

Figure 4: **Distribution of per-seed ρ_b across the transition** on the adaptive network (128×128, 64 seeds, 30k frames). At $\mu = 0.355$, the distribution is clearly bimodal with an empty gap between ordered and disordered basins. Vertical dashed lines mark the mean. Some seeds near the transition may undergo basin switching during the averaging window (cf. Sec. S3).

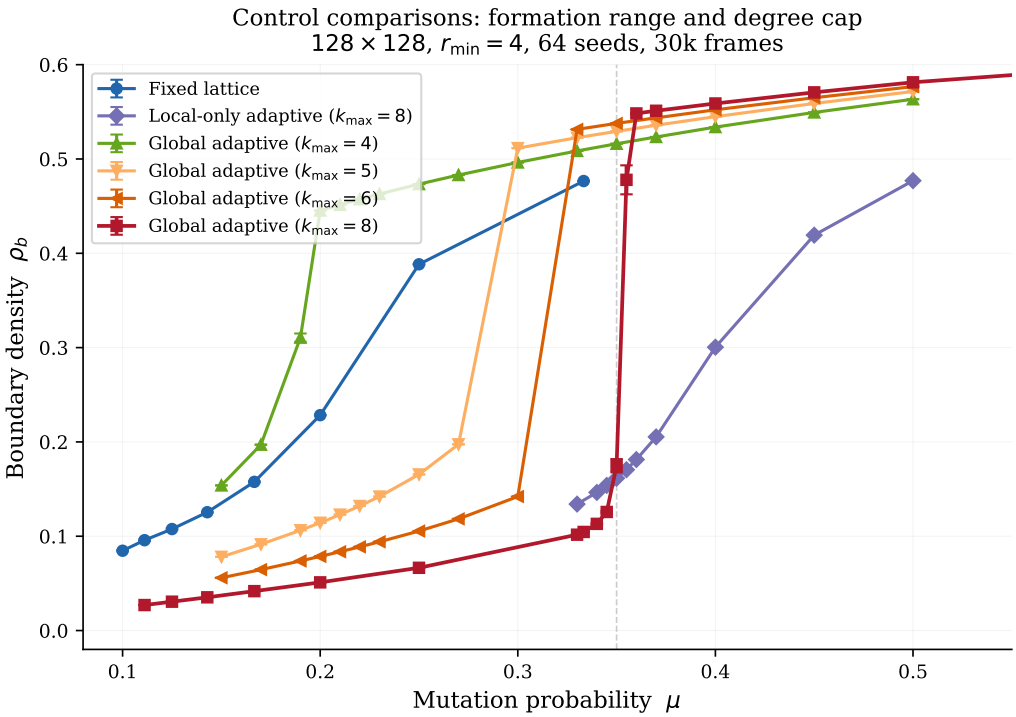

Figure 5: **Control comparisons: formation range and degree cap** (128×128, $r_{\min} = 4$, 64 seeds, 30k frames). Fixed lattice (blue), local-only adaptive with $k_{\max} = 8$ (purple), and global adaptive at four degree caps $k_{\max} \in \{4, 5, 6, 8\}$ (green, light orange, dark orange, red). The sharp coexistence-like jump is absent at $k_{\max} = 4$ but present at every $k_{\max} \geq 5$; the jump location shifts to higher μ with increasing headroom. Caveats: at $k_{\max} = 4$ the mean ordered-phase degree drops below $r_{\min}$, so reinforcement firing is suppressed by construction (see Section 6.1); the local-only control shrinks the candidate pool and so does not cleanly isolate non-locality.

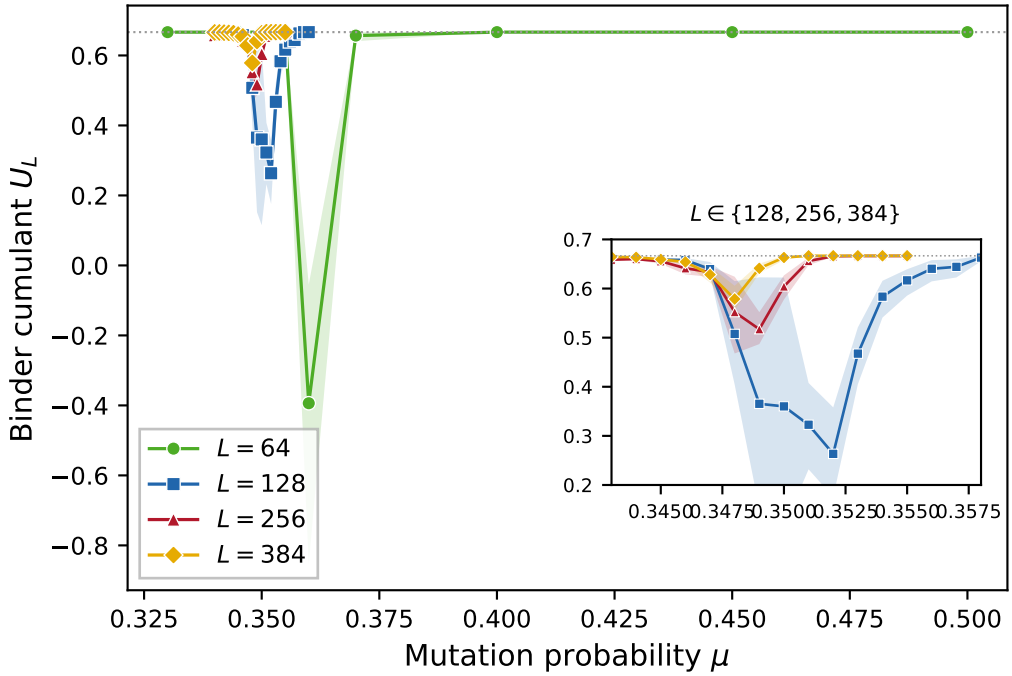

Figure 6: **Binder cumulant U_L vs. μ** for $L = 64$, 128, 256, and 384 on the adaptive network (fine μ grids, $\Delta\mu = 0.001$). The dip shifts to lower μ with increasing L (pseudo-critical drift) and shallows monotonically toward 2/3, consistent with a sharp continuous transition over the tested size range.